Advantages of rainwater harvesting

Rainwater harvesting helps to

(i) Overcome the rapid depletion of ground water levels.

(ii) To Meet the increase demand of water.

(iii) Reduces flood and soil erosion

(iv) Water stored in ground is not contaminated by human and animal wastes and hence can be used for drinking purpose.

22.8 Electrical Energy Management

Electricity or electric power is produced by generators. The generators are operated by the turbines attached to it. The turbines are rotated by steam, moving water or wind power to produce electricity.

Conservation of electrical energy

The following measures can be taken even at home and school to save electricity

(i) Use energy efficient appliances to save electricity like Compact Fluorescent Lamps (CFL), Light Emiting Diode (LED) bulbs and other electric equipments.

(ii) Switch off the lights and fans, television and other electrical appliances when not in use.

(iii) Switch of the mobile phone chargers when not in use.

(iv) Maximise the use of solar radiation. Solar water heating system can be used instead of electric geysers.

(v) Minimise the use of air conditioners.

22.9 E-Wastes and its Management

E-wastes are generally called as **electronic wastes**, which includes the spoiled, outdated, non-repairable electrical and electronic devices. These wastes contain toxic metals like lead, cadmium, chromium and mercury, though also contain iron, copper, silicon, aluminum and gold which can be recovered. Neverthless, only 5 % of e-wastes produced are recycled.

Sources of e-wastes

Electronic devices: Computers, laptops, mobile phones, printers, monitors, televisions, DVD players, calculators, toys, sport equipments, etc.

Household electrical appliances: Refrigerators, washing machine, microwave oven, mixer, grinder, water heater, etc.

Accessories: Printing cartridges, batteries and chargers.

E-wastes include

Computer components	- 66%
Telecommunication components	- 12 %
Electronic components	- 5 %
Biomedical components	- 7 %
Other components	- 6 %

Environmental impact of e-wastes

Disposal of any kind of electrical and electronic devices without knowledge can become the landfill and water pollutants.

More to Know

Health Effects of E- Wastes

Lead: Damages central and peripheral nervous system; affect brain development in children

Chromium: Asthmatic bronchitis

Cadmium: Accumulates in kidney and liver; neural damage

Mercury: Chronic damage to brain and respiratory system

Plastics including Polyvinyl Chloride (PVC): Burning produces dioxin which can cause developmental and reproductive problems, damages the immuns system.

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